

MATH 340 Examination

Multivariable Calculus and Vector Analysis

Instructions:

- Answer all questions.
 - For Questions 1–4, choose the best option.
 - For Questions 5–8, write brief but precise answers.
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1. What geometric property does the gradient vector ∇f at a point represent for a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$?
 - (A) The direction of maximum decrease of f
 - (B) A vector tangent to the level set of f at that point
 - (C) The direction of steepest ascent and perpendicular to level sets
 - (D) The curvature of the surface defined by f
 2. Which of the following conditions is sufficient for a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ to have its mixed partial derivatives equal, i.e., $f_{xy} = f_{yx}$?
 - (A) f is differentiable at the point
 - (B) f has continuous first partial derivatives
 - (C) f has continuous second partial derivatives
 - (D) f is twice differentiable at the point
 3. The Jacobian determinant of a transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ at a point provides information about:
 - (A) The rate of change of T along coordinate axes
 - (B) The local scaling factor of volumes under the transformation
 - (C) Whether T is injective globally
 - (D) The total derivative matrix of T
 4. Which statement correctly describes the relationship between Green's Theorem and Stokes' Theorem?
 - (A) Green's Theorem is a special case of Stokes' Theorem in two dimensions
 - (B) Stokes' Theorem generalizes the Divergence Theorem to higher dimensions
 - (C) Green's Theorem applies to simply connected regions only

- (D) They are independent theorems with no direct relationship
5. Explain the geometric and computational significance of the Hessian matrix in analyzing critical points of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. How does the second derivative test for functions of several variables use the Hessian?
 6. Describe the difference between a conservative vector field and a path-independent vector field. Under what conditions are these properties equivalent, and what role does the curl play?
 7. State the Divergence Theorem (Gauss's Theorem) and explain its physical interpretation in the context of fluid flow. How does it relate local behavior (divergence) to global behavior (flux)?
 8. Explain why changing the order of integration in a double integral requires careful consideration of the region of integration. Provide a method for determining the new limits when switching from $\int \int dx dy$ to $\int \int dy dx$.